ICE-4107: Microwave Engineering

Lecture 3

Electromagnetic Plane Waves

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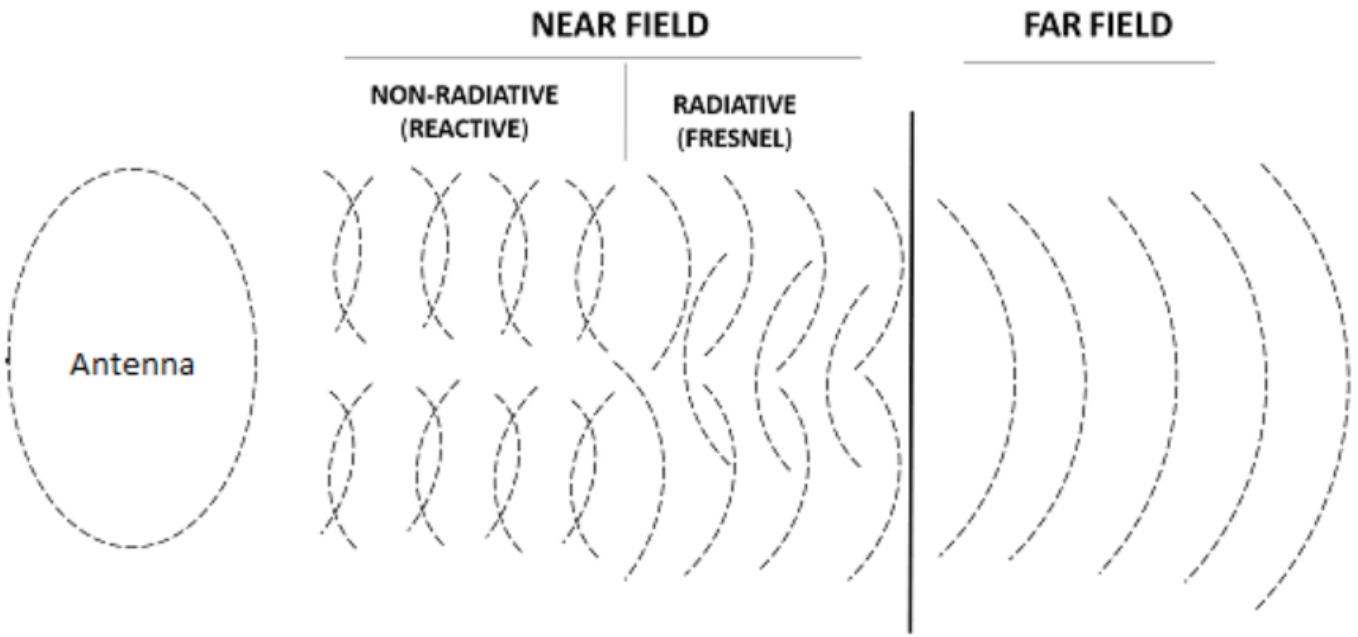
Introduction

- The principles of electromagnetic plane waves are based on the relationships between electricity and magnetism.
- A changing magnetic field will induce an electric field, and a changing electric field will induce a magnetic field.
- Induced fields are not confined but ordinarily extend outward into space.
- The sinusoidal form of the wave causes energy to be interchanged between the magnetic and electric fields in the direction of the wave propagation.

Introduction

- In the far field of free space, electric and magnetic waves are always perpendicular to each other, and both are normal to the direction of propagation of the wave. This type of wave is known as the transverse electromagnetic (TEM) wave.

- If only the transverse electric wave exists, the wave is called TE-mode wave. That means there is no component of the electric wave in the direction of propagation. In TM modes only the transverse magnetic wave exists.



The EM field characteristics vary as a function of distance from the antenna. They are broadly divided into two regions, the near-field region, and the far field region.

The **Near Field Region** is the region right next to the antenna. In this region, the fields are sort of unpredictable and therefore no measurements are usually made in this region. It is defined by the following equation:

$$Near\ Field\ Region < \frac{2D^2}{\lambda}$$

Where D = Maximum linear dimension of the antenna

λ = Wavelength of the EM Waves

Near-Field region is further divided into two parts:

Reactive Near Field: This is the region that is adjacent to the antenna. In this region, the E-Field and H-Field are 90 degrees out of phase with each other and are therefore reactive. To radiate or propagate the E/H fields need to be orthogonal (perpendicular) and in phase with each other.

$$Reactive\ Near\ Field\ Region\ <\ 0.62\sqrt{\frac{D^3}{\lambda}}$$

Where D = Maximum linear dimension of the antenna

λ = Wavelength of the EM Waves

Radiative Near Field: This region is also known as the Fresnel Region. It is the region between the reactive near field and the far field. This is the region where the EM fields start to transition from reactive to radiating fields. However, since they have not completely transitioned, the shape of the radiation pattern still varies with distance.

$$0.62\sqrt{\frac{D^3}{\lambda}} < \textbf{\textit{Radiative Near Field Region}} < \frac{2D^2}{\lambda}$$

Where D = Maximum linear dimension of the antenna

λ = Wavelength of the EM Waves

The **Far Field Region** is the region that comes after the near radiative near field. In this region, the EM fields are dominated by radiating fields. The E and H-fields are orthogonal to each other and to the direction of propagation as with plane waves. Antennas are usually used to transfer signals at large distances which are considered to be in the far-field region. One condition that must be met when making measurements in the far field region is that the distance from the antenna must be much greater than the size of the antenna and the wavelength. The far-field region is represented by the following equation:

$$\textit{Far Field Region} > \frac{2D^2}{\lambda}$$

Where D = Maximum linear dimension of the antenna

λ = Wavelength of the EM Waves

Maxwell's Equation

Maxwell's Equation describe all electromagnetic phenomena. The equations are given below:

$$\nabla \times \boldsymbol{E} = -\frac{\partial \boldsymbol{B}}{\partial t}$$

$$\nabla \times \boldsymbol{H} = \boldsymbol{J} + \frac{\partial \boldsymbol{D}}{\partial t}$$

$$\nabla \cdot \boldsymbol{D} = \rho$$

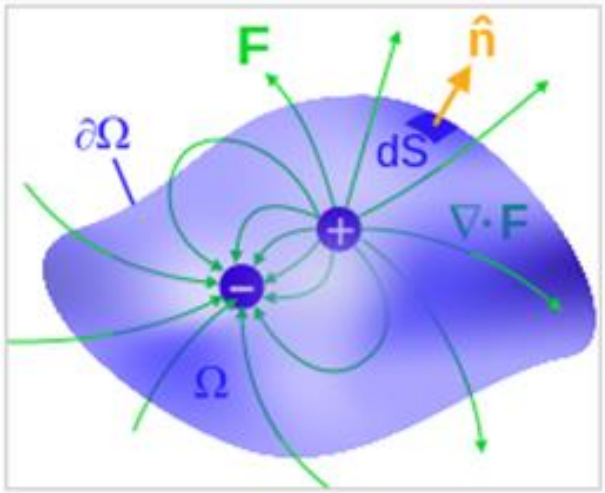
$$\nabla \cdot \boldsymbol{B} = 0$$

The first is Faraday's law of induction, the second is Ampere's law as amended by Maxwell to include the displacement current $\partial \boldsymbol{D}/\partial$t, the third and fourth are Gauss' laws for the electric and magnetic fields. The right-hand side of 4th equation is zero because there is no magnetic monopole charges.

Ω = Omega

Flux and Divergence

- The fields emanating from the sources can be inferred from the surface integrals of the fields through the closed surface $\partial\Omega$ denoted as the electric flux $\oiint_{\partial\Omega} \mathbf{E} \cdot d\mathbf{S}$ and magnetic flux $\oiint_{\partial\Omega} \mathbf{B} \cdot d\mathbf{S}$ as well as their respective divergences .
- These surface integrals and divergences are connected by the divergence theorem



- **Maxwell's Equation**

Name	Differential equations	Meaning
Gauss's law	$\nabla \cdot \mathbf{E} = \frac{\rho}{\varepsilon_0}$	The electric field leaving a volume is proportional to the charge inside.
Gauss's law for magnetism	$\nabla \cdot \mathbf{B} = 0$	There are no magnetic monopoles; the total magnetic flux piercing a closed surface is zero.
Maxwell–Faraday equation (Faraday's law of induction)	$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}$	The voltage accumulated around a closed circuit is proportional to the time rate of change of the magnetic flux it encloses.
Ampère's circuital law (with Maxwell's addition)	$\nabla \times \mathbf{B} = \mu_0\left(\mathbf{J} + \varepsilon_0\frac{\partial \mathbf{E}}{\partial t}\right)$	Electric currents and changes in electric fields are proportional to the magnetic field circulating about the area they pierce.

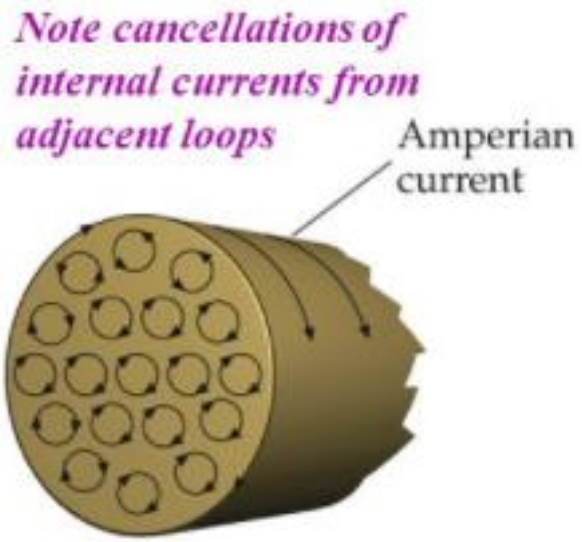
[Note: Lenz's law is shown with the negative sign in Faraday's law of induction: $\mathcal{E} = -\frac{\partial \Phi}{\partial t}$, which indicates that the induced voltage and the change in magnetic flux have opposite signs.]

Note on free current vs. bound current

- The electric current that passes through a wire or battery would be classified as "free current“.
- In contrast, "bound current" arises in the context of bulk materials that can be magnetized and/or polarized.
- When a material is magnetized (for example, by placing it in an external magnetic field), the electrons remain bound to their respective atoms, but behave as if they were orbiting the nucleus in a particular direction, creating a microscopic current.

Note on free current vs. bound current

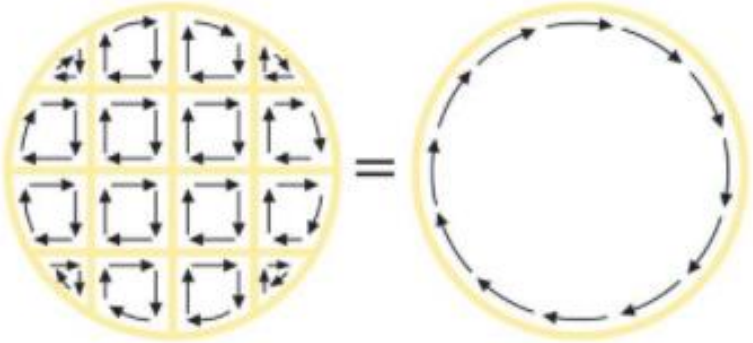
- When the currents from all these atoms are put together, they create the same effect as a macroscopic current, circulating continually around the magnetized object.
- This magnetization current $\mathbf{J}_M$ is one contribution to "bound current".

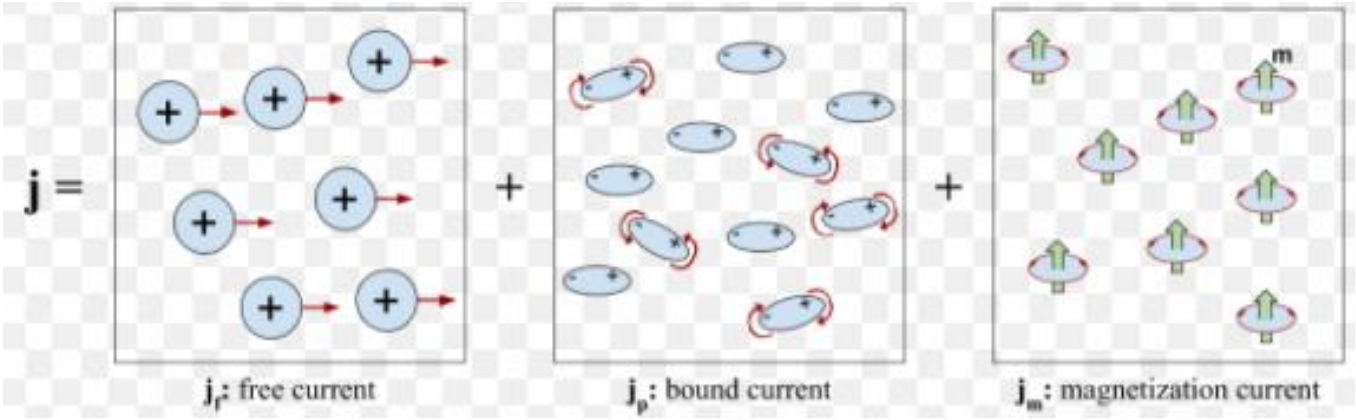


Note on free current vs. bound current

- Net current inside the material is zero. We will have surface current and thus a magnetic moment.
- The other source of bound current is bound charge.
- When an electric field is applied, the positive and negative bound charges can separate over atomic distances in polarizable materials, and when the bound charges move, the polarization changes, creating another contribution to the "bound current", the polarization current $\mathbf{J}_P$.

- **Note on free current vs. bound current**





Note on free current vs. bound current

- The total current density **J** due to free and bound charges is then:

$$\mathbf{J} = \mathbf{J_f} + \mathbf{J_M} + \mathbf{J_P}$$

with $\mathbf{J}_f$ the "free" or "conduction" current density.

❑ Displacement Current

- In free space, the displacement current ($\varepsilon_0 \partial E/\partial t$) is related to the time rate of change of electric field.
- However, in a dielectric the above contribution to displacement current is present too, but a major contribution to the displacement current is related to the polarization of the individual molecules of the dielectric material.

Displacement Current

- Even though charges cannot flow freely in a dielectric, the charges in molecules can move a little under the influence of an electric field.
- The positive and negative charges in molecules separate under the applied field, causing an increase in the state of polarization, expressed as the polarization density **P**.
- A changing state of polarization is equivalent to a current.

Displacement Current

- Thus, in the expression for displacement current, it has two components:

$$\mathbf{J}_\mathrm{D} = \varepsilon_0 \frac{\partial \mathbf{E}}{\partial t} + \frac{\partial \mathbf{P}}{\partial t}.$$

- The first term on the right-hand side is present everywhere, even in a vacuum. It doesn't involve any actual movement of charge, but it nevertheless has an associated magnetic field, as if it were an actual current.
- The second term on the right-hand side is the displacement current as originally conceived by Maxwell, associated with the polarization of the individual molecules of the dielectric material.

The Maxwell–Ampère equation

- Treating all charges, the generalized Ampère's equation, also called the Maxwell–Ampère equation, is in differential form

$$\nabla \times \mathbf{B} = \mu_0 \mathbf{J} + \mu_0 \epsilon_0 \frac{\partial \mathbf{E}}{\partial t}$$

- In both forms **J** includes magnetization current density as well as conduction and polarization current densities. That is, the current density on the right side of the Ampère–Maxwell equation is:

$$\mathbf{J}_\mathrm{f} + \mathbf{J}_\mathrm{D} + \mathbf{J}_\mathrm{M} = \mathbf{J}_\mathrm{f} + \mathbf{J}_\mathrm{P} + \mathbf{J}_\mathrm{M} + \varepsilon_0 \frac{\partial \mathbf{E}}{\partial t} = \mathbf{J} + \varepsilon_0 \frac{\partial \mathbf{E}}{\partial t}\,,$$

where current density $\mathbf{J}_\mathrm{D}$ is the displacement current, and **J** is the current density contribution actually due to movement of charges, both free and bound

Maxwell's Equations: Applications

- **Prediction of Electromagnetic Waves**

Maxwell mathematically manipulated his equations (uncoupling the partial differential equations, applying the curl etc.) and derived a wave equation.

- **Speed of Propagation of Electromagnetic Waves**

Maxwell deduced the speed of these electromagnetic waves theoretically to 3×10^8 m/sec.

$$\frac{1}{(\mu_0\epsilon_0)^{1/2}} = 3\times10^8 \text{ meters/ sec}$$

He observed that the speed of propagation of electromagnetic waves was very close to the speed of light.

Maxwell's Equations: Applications

Maxwell thereby demonstrated that light was a type of electromagnetic wave. He published a theory that accounted for the physical origins of light. Thus, he unified the fields of electromagnetism and optics, which were unrelated separate branches until then.

- **Theory of Relativity**

Maxwell's equations have a close relation to special relativity. Albert Einstein himself said: *"The special theory of relativity owes its origins to Maxwell's equations of the electromagnetic field."* The simplicity, symmetry and beauty of Maxwell's equations motivated Einstein to develop the revolutionary theory of relativity.

Maxwell's Equations: Applications

i. To describe many theories: Poynting theorem, Lorentz potential,

ii. To describe transmission line, waveguide, strip line, antenna, etc.

iii. Magnetic train, Biomedical instruments (MRI, Imaging, etc.), Magnetic tape

iv. Any devices that use electricity or magnets.

Thank You